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Course Name : Data analytics with GenAi

Assignment :Calculated Fields, Parameters & LOD Expressions

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**Question 1**
**What is a Calculated Field and why is it important in business dashboards?**
**Answer**

A **Calculated Field** in Tableau is a custom field created by users using formulas, functions, and existing fields to perform calculations that are not available in the original data source. It allows you to create new metrics, such as profit ratio, growth percentage, or conditional flags.

Calculated Fields are extremely important in business dashboards because they enable deeper data analysis, custom KPI creation, and dynamic insights without modifying the original dataset. They make dashboards more flexible, interactive, and meaningful for decision-making by transforming raw data into actionable business intelligence.

**Question2**
**Why can't parameters change visuals directly in Tableau?**
**Answer**

Parameters in Tableau cannot change visuals directly because they are **user-input controls** that only pass a single value (string, number, date, or boolean) to the workbook. They act like variables but do not contain any logic or instructions on how to modify the view.

To actually change visuals, parameters must be used inside **Calculated Fields**. The calculated field then applies conditional logic (using CASE, IF statements) to decide what to show or how to filter the data. Only then can this calculated field control the visualization, colors, filters, or measures.

Thus, parameters are powerful but need calculated fields as a bridge to impact visuals.

**Question3**
**When should EXCLUDE LOD be used?**
**Answer**

**EXCLUDE LOD** should be used when you want to **remove (ignore) one or more dimensions** present in the view for a specific calculation, while keeping the detailed view intact.

It is ideal for:

- Calculating **Percent of Total** (e.g., sales contribution ignoring a lower dimension like City or Product)
- Finding **difference from overall average**
- Comparing individual values against a higher-level total (like Region total while viewing by State and City)

EXCLUDE computes at a **coarser (higher)** level of detail than the view. It is useful when you need subtotals or reference values without changing the visual granularity.

**Question 4 : Difference between FIXED and INCLUDE LOD expressions?**

The main difference between FIXED and INCLUDE LOD expressions in Tableau lies in how they control the level of detail in calculations:

- FIXED LOD: Ignores all dimensions in the view and computes at a specific level defined in the expression. It returns the same value for every row within the same FIXED dimension(s), regardless of filters or other dimensions in the view. It is useful for creating consistent aggregates like total sales by Region even when viewing by City.
- INCLUDE LOD: Adds extra dimensions to the view's level of detail for the calculation. It computes at a finer (lower) granularity than the view and then aggregates up. It is ideal when you want more detailed calculations (e.g., average of monthly sales while viewing yearly data).

**practical**
[**Calculated Fields, Parameters2.twb**]